

Cambridge (CIE) IGCSE Chemistry
Paper 1: Multiple Choice (Set A)
Core

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

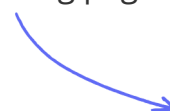
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 Many chemical processes involve substances in either the liquid or the gaseous state.

Which of the following statements is correct?

- A. A specific mass of a gas has a fixed volume at room temperature
- B. A specific mass of a liquid has a fixed volume at room temperature
- C. 25 cm^3 of gas contains more particles than 25 cm^3 of liquid
- D. Liquid particles placed inside a sealed container have fixed positions

(1 mark)

- 2 The following four statements describe the arrangement of particles in a substance.

1	Particles are moving in all directions
2	Particles are arranged in neat rows
3	Particles can move past each other
4	Particles vibrate about their positions

Which of the four statements describe the arrangement of particles in a solid?

- A. 1 and 2
- B. 2 and 4
- C. 1 and 3
- D. 2 only

(1 mark)

- 3 The structure of four particles is described in the table.

Particle	Number of protons	Number of neutrons	Number of electrons
Li	3	X	3
Li ⁺	3	4	2
F	Y	10	9
F ⁻	9	10	Z

What are the correct values for X, Y and Z?

	X	Y	Z
A	4	7	9
B	3	9	9
C	4	9	10
D	3	7	10

(1 mark)

4 Ionic compounds are formed when a metal reacts with a nonmetal.

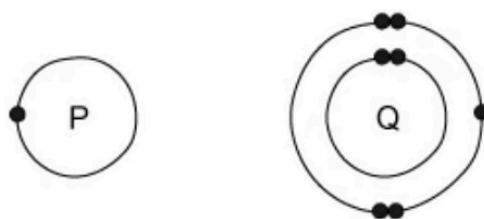
Which two elements combine to form this kind of compound?

Element	Electronic structure
Q	2.4
R	2.8
S	2.8.1
T	2.8.7

- A. S and T
- B. R and S
- C. Q and R
- D. Q and T

(1 mark)

- 5 Atoms P and Q have electronic structures as shown in the diagram. They react together to form a covalent compound.



What is the formula of the compound?

- A. P_2Q_3
- B. P_3Q
- C. P_3Q_2
- D. PQ_4

(1 mark)

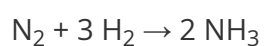
6 Which of the following statements about graphite and diamond are incorrect?

1	They are allotropes of carbon.
2	They both conduct electricity.
3	They form different numbers of bonds within their structures.
4	They have different uses.

- A. 2 and 3
- B. 2 only
- C. 1, 3 and 4
- D. 2 and 4

(1 mark)

7 Ammonia is manufactured by the reaction of hydrogen and nitrogen in the Haber process, as shown in the equation below:



16.8 tonnes of ammonia can be produced from 14 tonnes of nitrogen.

How much nitrogen is needed to produce 51 tonnes of ammonia?

- A. 42.5
- B. 34
- C. 17
- D. 24.5

(1 mark)

8 Dilute sulfuric acid is electrolysed using inert electrodes.

What are the products of this electrolysis?

	cathode	anode
A	oxygen	sulfur
B	hydrogen	sulfur dioxide
C	oxygen	hydrogen
D	hydrogen	oxygen

(1 mark)

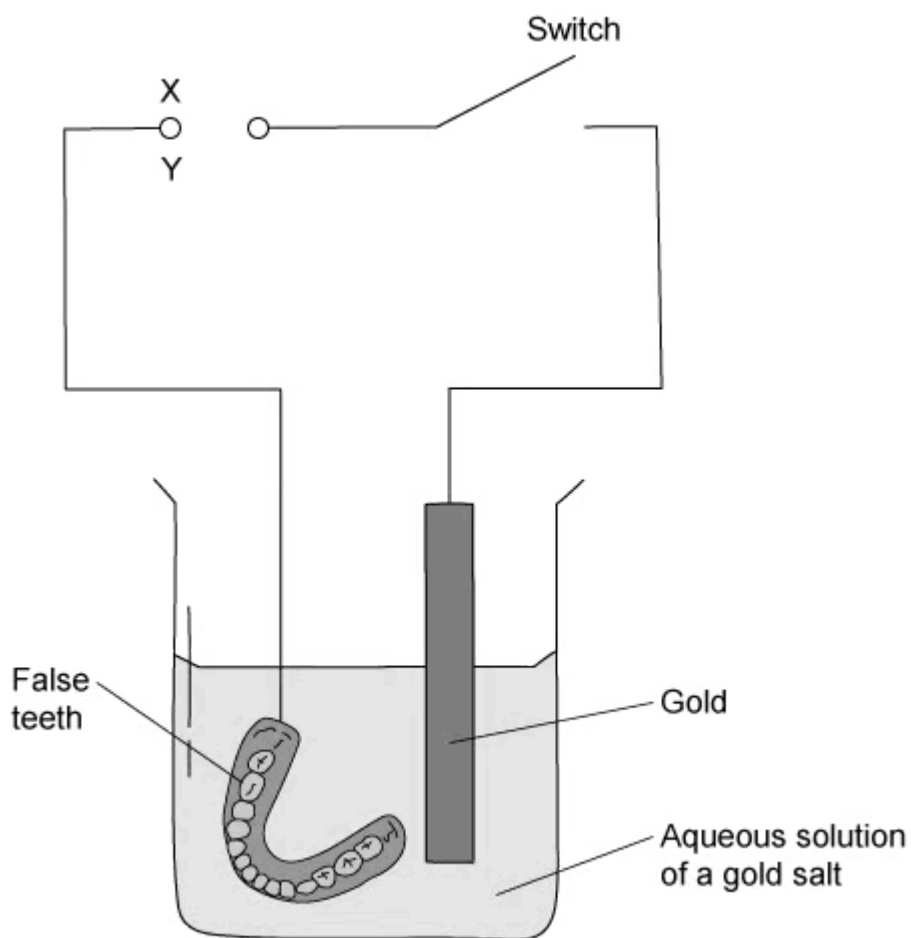
9 Which metal could **not** be used for electroplating by using an aqueous solution?

- A. Chromium
- B. Copper
- C. Silver
- D. Sodium

(1 mark)

10 Winston Churchill, a British Prime Minister, had his false teeth electroplated with gold.

The teeth were coated with a thin layer of carbon and were then placed in the apparatus shown.



	terminal X is	the carbon powder could be
A	negative	diamond
B	negative	graphite
C	positive	diamond
D	positive	graphite

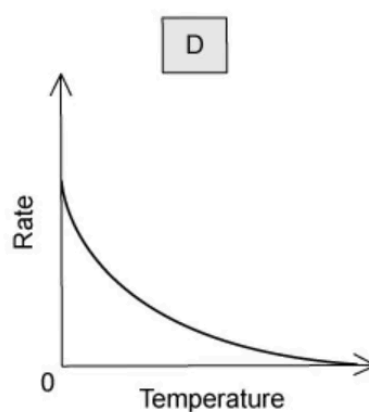
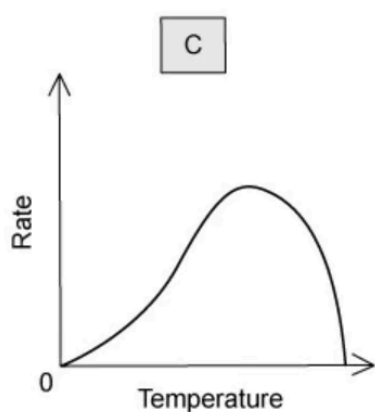
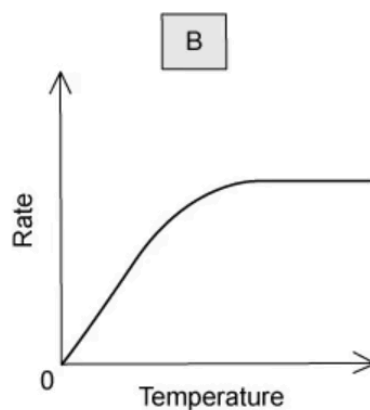
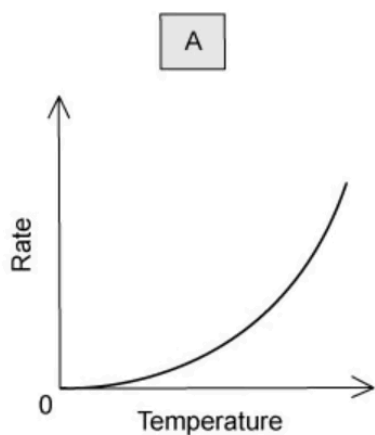
(1 mark)

11 Which of the following processes is endothermic?

- A. Reacting sodium with water.
- B. The use of petrol in an engine.
- C. Distilling crude oil.
- D. Burning fossil fuels.

(1 mark)

12 Which graph correctly shows the effect of increasing the temperature on the rate of reaction between magnesium and sulfuric acid in which neither reactant runs out?



A. Option A

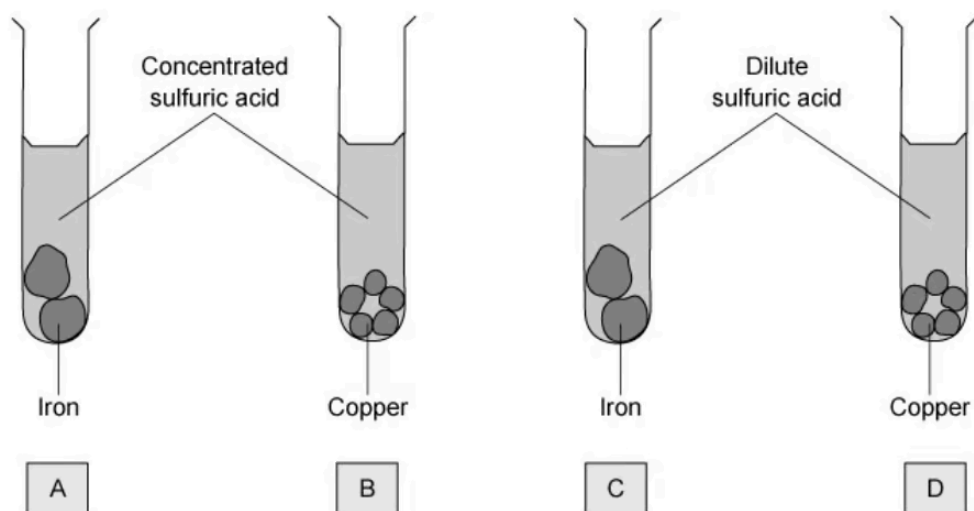
B. Option B

C. Option C

D. Option D

(1 mark)

- 13** A student was investigating the rate of reaction between acid and metals. The experimental set-up is shown below.



Which test tube would give the quickest reaction?

A. Option A

B. Option B

C. Option C

D. Option D

(1 mark)

- 14** A student wanted to prepare copper(II) sulfate crystals by reacting solid copper(II) oxide to a solution of hot sulfuric acid.

An extract from the method section of the experimental handout is shown below:

Add the copper(II) oxide in small amounts to the hot acid in the reaction flask and stir until it is**1**..... When the reaction has gone to completion the copper(II) oxide will no

longer**2**.....,**3**..... and**4**..... the solution to obtain the copper(II) sulfate crystals.

	1	2	3	4
A	in excess	precipitate	cool	dissolve
B	in excess	dissolve	filter	cool
C	reacting	dissolve	cool	filter
D	reacting	precipitate	filter	cool

(1 mark)

15 Three reactions are shown below.

- $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$
- $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$
- $2\text{MgO} + \text{O}_2 \rightarrow 2\text{MgO}_2$

Which row correctly describes the process each reactant molecule undergoes?

	NO	O₃	MgO
A	oxidised	reduced	oxidised
B	oxidised	oxidised	reduced
C	reduced	oxidised	oxidised
D	reduced	reduced	oxidised

(1 mark)

- 16** A student is provided with an unmarked beaker containing a colourless solution and asked to identify it. The student decides to perform a series of reactions with the solution and record the observations.

Which of these observations is not indicative of an acid?

- A.** The solution causes blue litmus to turn red.
- B.** On heating and the addition of copper oxide, the solution turns blue.
- C.** On the addition of ammonium carbonate, a colourless, pungent gas is released.
- D.** On the addition of a piece of magnesium, effervescence was produced and the magnesium piece 'disappeared' after a few minutes.

(1 mark)

- 17** Which indicator displays the correct colour in each solution?

Indicator	Colour in dilute sulfuric acid	Colour in potassium hydroxide solution
methyl orange	red	yellow
universal indicator	red	orange
litmus	red	purple
thymolphthalein	blue	colourless

- A.** Methyl orange
- B.** Universal indicator
- C.** Litmus
- D.** Thymolphthalein

(1 mark)

- 18** The early Periodic table was used to classify the known elements and was used to predict the properties of the elements that had not yet been discovered.

Which row correctly describes how this was possible?

- A.** Elements in the same Period displayed similar properties.
- B.** The arrangement of the known elements provided trends in behaviour which was used to make predictions about the unknown elements.
- C.** Elements in different Groups displayed similar properties.
- D.** The unknown elements were used to fill the gaps in the original Periodic table and were placed according to their physical properties.

(1 mark)

19 Which statement about the Alkali metals is correct?

- A.** Potassium is more dense than lithium.
- B.** Lithium has a higher density than sodium.
- C.** Potassium is less reactive than sodium.
- D.** The melting point of sodium is higher than lithium.

(1 mark)

20 Which row describes the properties of a typical noble gas?

	outer electrons	atom or molecule	boiling point
A	full	atom	very low
B	6	molecule	low
C	8	atom	very high
D	6	molecule	high

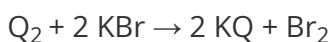
(1 mark)

21 The reactivity of the alkali metals increases moving down the group. Which statement explains this observation?

- A.** Decreasing distance between the nucleus and outer shell makes it easier for the metals to lose their valence electrons.
- B.** Increasing distance between the nucleus and outer shell makes it harder for the metals to lose their valence electrons.
- C.** Increasing distance between the nucleus and outer shell makes it easier for the metals to lose their valence electrons.
- D.** Decreasing distance between the nucleus and outer shell makes it easier for the metals to gain their valence electrons.

(1 mark)

22 Element Q is a halogen which reacts with aqueous potassium bromide, as shown in the reaction below:

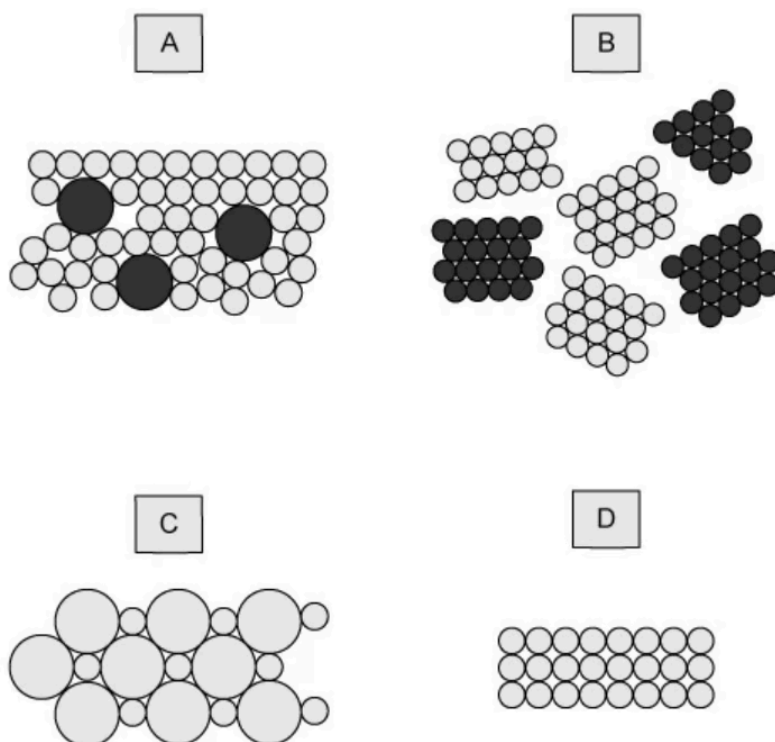


Which row describes element Q?

	More reactive than bromine	Greater A_r than bromine
A	✓	✓
B	✓	×
C	×	✓
D	×	×

(1 mark)

23 Which diagram shows an alloy?



(1 mark)

24 A sample of element Y is shiny and can be shaped by hammering.

Which row of the table corresponds to the properties of element Y?

	conducts electricity	melts below 25°C
A	×	✓
B	×	×
C	✓	✓
D	✓	×

A. Option A

B. Option B

C. Option C

D. Option D

(1 mark)

25 The three statements below about aluminium are all correct.

1	Aluminium reacts readily to form aluminium oxide
2	Aluminium is extracted from its ore by electrolysis as it cannot be extracted by reduction with carbon
3	Aluminium is used in electrical wiring because of its good electrical conductivity

In which of these statements can '**aluminium**' be replaced by '**copper**' and still be correct?

	1	2	3
A	✓	✓	✓
B	x	✓	x
C	✓	x	✓
D	x	x	✓

A. Option A

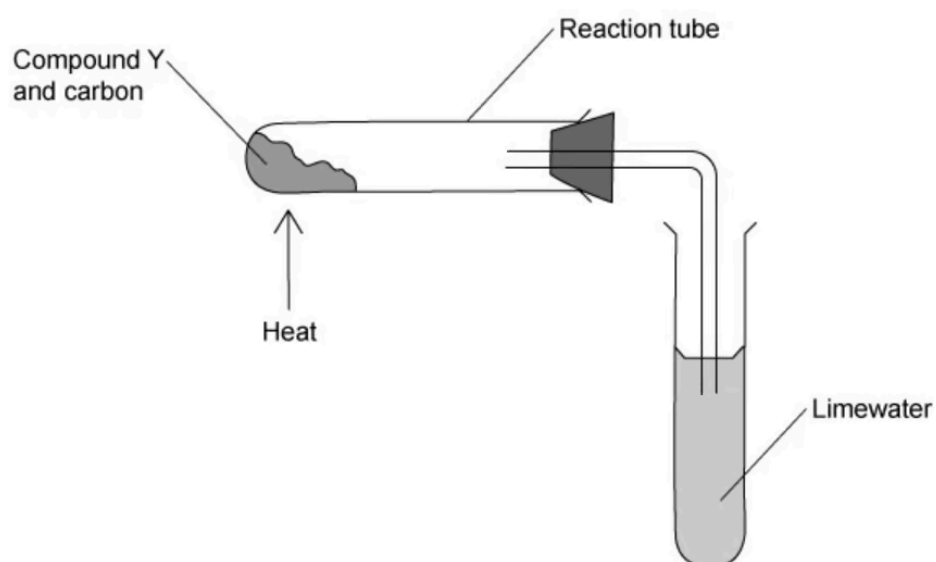
B. Option B

C. Option C

D. Option D

(1 mark)

26 Compound Y is thermally decomposed with carbon using the apparatus below:



A brown solid is formed in the reaction tube and the limewater turns cloudy.

What is compound Y?

- A.** Copper (II) oxide
- B.** Calcium oxide
- C.** Magnesium oxide
- D.** Sodium oxide

(1 mark)

27 **Q**, **R** and **S** are metals.

Metal	Reacts with cold water	Reacts with steam	Reacts with dilute hydrochloric acid
Q	yes	yes	violently
R	very slowly	slowly	less strongly
S	no	no	no

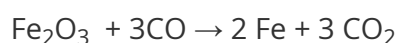
What are the possible identities of the metals **Q**, **R** and **S**?

	Q	R	S
A	Potassium	Iron	Magnesium
B	Sodium	Zinc	Copper
C	Silver	Aluminium	Iron
D	Calcium	Copper	Gold

- A. Option A
- B. Option B
- C. Option C
- D. Option D

(1 mark)

- 28 In a blast furnace, iron (III) oxide is converted into iron and carbon monoxide is converted into carbon dioxide, as shown in the equation:



What happens to each of these reactants?

- A. Iron (III) oxide is reduced and carbon monoxide is oxidised.
- B. Iron (III) oxide is oxidised and carbon monoxide is reduced.
- C. Both iron (III) oxide and carbon monoxide are oxidised.
- D. Both iron (III) oxide and carbon monoxide are reduced.

(1 mark)

- 29 The three statements about aluminium are all correct.

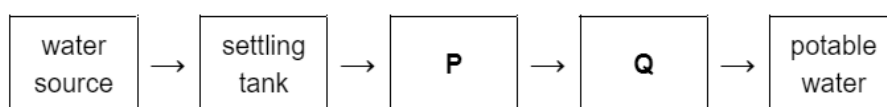
1	Aluminium has an oxide ore.
2	Aluminium is extracted from its ore by electrolysis.
3	Aluminium is used to make cooking utensils.

Which of the statements would still be correct if aluminium was replaced with iron?

	1	2	3
A	×	✓	✓
B	✓	×	✓
C	✓	✓	✓
D	✓	×	×

(1 mark)

- 30 Potable water is produced from a water source near a country spring as shown in the flow diagram.



Which processes occur at P and Q?

	P	Q
A	chlorination	filtration
B	chlorination	distillation
C	filtration	distillation
D	filtration	chlorination

(1 mark)

- 31** Four chemical processes are described below. Which reactions contribute to global warming?

1	Reacting hydrochloric acid with magnesium carbonate.
2	The complete combustion of propane.
3	Heating calcium carbonate.
4	Reacting sodium hydroxide with magnesium carbonate.

A. 1, 2 & 3

B. 1 & 4

C. 1, 3 & 4

D. All 4

(1 mark)

- 32** Which row correctly shows two ways of reducing the formation of acid rain?

A	Use a catalytic convertor	Reduce the amount of livestock farming
B	Plant more trees	Use fuels containing low levels of sulfur
C	Use fuels containing low levels of sulfur	Use a catalytic convertor
D	Reduce the amount of livestock farming	Plant more trees

(1 mark)

33 Flue gas desulfurisation uses calcium oxide to reduce the emissions of sulfur dioxide.

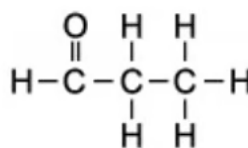
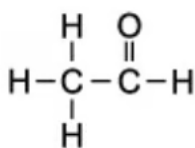
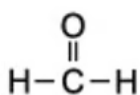
Which statements are correct?

1	Sulfur dioxide is a basic oxide
2	Sulfur dioxide is responsible for the production of acid rain
3	Calcium oxide reacts with acids
4	Sulfur dioxide emissions can also be reduced by using low sulfur fuels

- A.** 1, 2 and 3
B. 1 and 2 only
C. 1, 3 and 4
D. 2, 3 and 4

(1 mark)

34 The first three members of a homologous series are shown.

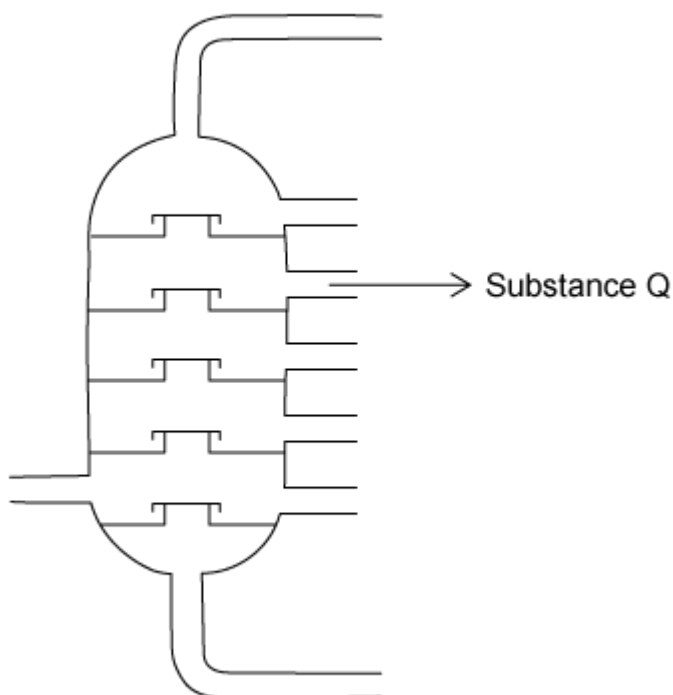


Why do these molecules represent a homologous series?

- A.** They contain carbon and hydrogen atoms.
B. They have saturated bonds.
C. They have the same functional group.
D. They react differently from each other.

(1 mark)

- 35** The diagram shows an industrial process involved in the production of a substance that is used as an aircraft fuel.



	process	substance Q
A	fractional distillation	petrol
B	thermal decomposition	paraffin
C	fractional distillation	paraffin
D	thermal decomposition	petrol

(1 mark)

- 36** The table below shows 4 different sources of petroleum (crude oil) and their composition.

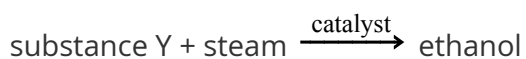
Fraction	Arabian Heavy /%	Arabian Light /%	Iranian Heavy /%	North Sea /%
gasoline	18	21	21	23
kerosene	11.5	13	13	15
diesel	18	20	20	24
fuel oil	52.5	46	46	38

Which composition of petroleum would be the best for the motor vehicle industry?

- A.** North Sea
- B.** Arabian Heavy
- C.** Arabian Light
- D.** Iranian Heavy

(1 mark)

- 37** Ethanol can be produced from substance Y.

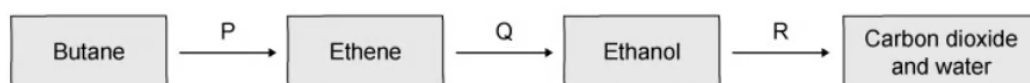


What is substance Y?

- A.** Ethane
- B.** Ethene
- C.** Carbon dioxide
- D.** Hydrogen

(1 mark)

38 The diagram shows a sequence of reactions.



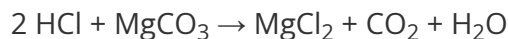
Which row correctly names the processes **P**, **Q** and **R**?

	P	Q	R
A	distillation	fermentation	respiration
B	distillation	hydration	combustion
C	cracking	fermentation	respiration
D	cracking	hydration	combustion

(1 mark)

- 39** A student wants to investigate the rate of reaction between marble chips and hydrochloric acid by measuring the volume of carbon dioxide produced with different concentrations of acid.

The equation for the reaction is as follows:



Which of the following lists of equipment can she use to measure the volume of the gas produced?

- A.** Gas syringe, cotton wool, bung, delivery tube
- B.** Gas syringe, conical flask, bung, delivery tube
- C.** Gas syringe, beaker, bung, balance
- D.** Balloon, conical flask, stopper, balance

(1 mark)

- 40** A substance was analysed and was found to have a boiling point of 450 °C.

At room temperature it is a soft grey solid.

Which temperature could be the melting point of a pure sample of the substance?

- A.** - 12 °C
- B.** 56 °C to 72 °C
- C.** 250 °C to 275 °C
- D.** 142 °C

(1 mark)